

**UNITED STATES OF AMERICA
BEFORE THE
FEDERAL ENERGY REGULATORY COMMISSION**

Interconnection of Large Loads to the Interstate
Transmission System

Docket No. RM26-4-000

REPLY COMMENTS OF CRUSOE ENERGY SYSTEMS, INC.

Pursuant to the Federal Energy Regulatory Commission's (Commission) October 27, 2025 Notice Inviting Comments and the November 7, 2025 Notice Granting Extension of Time, Crusoe Energy Systems, Inc. (Crusoe) submits these reply comments in support of the Department of Energy's (DOE) Advanced Notice of Proposed Rulemaking (ANOPR) requesting that FERC establish standardized interconnection procedures and agreements for large loads seeking to interconnect to the transmission system.¹

I. SUMMARY OF COMMENTS

Crusoe is a vertically integrated Artificial Intelligence (AI) infrastructure company that builds, owns, and operates data centers specifically optimized for the high-performance computing requirements of artificial intelligence and machine learning workloads. Crusoe's primary operations involve sourcing and developing cost-effective energy, including natural gas, intermittent renewable power (*e.g.*, wind, solar) and battery storage, to power its data center facilities. Crusoe leverages its in-house manufacturing capabilities to rapidly deploy computing infrastructure that provides reliable, low-latency cloud services, including access to leading-edge graphics processing units (GPUs), to enterprise customers and AI developers.

¹ Letter from DOE Secretary Wright to the Federal Energy Regulatory Commission (Oct. 23, 2025), attaching the Advanced Notice of Proposed Rulemaking, *Ensuring Timely and Orderly Interconnection of Large Loads*.

Through its energy-first approach, Crusoe aims to reduce both the cost and the environmental impact of utility-scale computing infrastructure.

The proposed technical requirements discussed in the ANOPR could enable companies like Crusoe that control both the build and design specifications of their facilities to serve as effective early implementers of these enhanced interconnection standards. Crusoe is actively contributing to the development of technical and policy frameworks across the country that enhance grid reliability while accommodating flexible load operations.

To prevent market inefficiencies, the Commission should establish heightened standards for large load and hybrid facility interconnection requests, including a significant increase in deposit fees and site control requirements to deter speculative requests. Additionally, large load and hybrid facility interconnection customers must be required to demonstrate actual technical capability and financial commitment, rather than securing queue positions based solely on qualitative commitments that could impede legitimate development opportunities. Without such safeguards, speculative queue positions could significantly disrupt AI computing development by creating uncertainty around the transferability and value of interconnection rights and potentially deterring investment in data center development.

The Commission should also prioritize hybrid facilities that combine load interconnection requests with onsite generation because they can reduce load requirements and provide valuable generation support to the grid during peak demand. Hybrid facilities should be studied based on the specific load withdrawal and generation injection rights requested and be recognized as structures that cause significantly less impact to the transmission grid than large loads that do not bring generation.

Finally, reforms are needed on the federal level to support the rapid deployment of AI infrastructure. Coordinated federal action is needed to improve transmission construction and equipment procurement timelines. In addition, the Commission should adopt Public Utility Holding Company Act (PUHCA) regulation reforms to remove federal impediments to power supply arrangements by eliminating “sleeve” power purchase agreement (PPA) requirements where retail sales are allowed by states.

II. REPLY COMMENTS

A. Reply Comments on the ANOPR’s Principles for Reform

Principle 1: Large load and hybrid facility interconnection processes should be developed pursuant to federal jurisdiction.

Large load and hybrid facility interconnections will have a significant impact on the regional and interregional transmission grid across multiple states. Most commenters agree that large load and hybrid facility interconnection processes should be developed pursuant to federal jurisdiction to ensure relatively consistent rules apply across the country.

Principle 2: Reforms should be applicable to new loads greater than 100 MW.

The proposed 20 MW threshold for the new large load and hybrid facility interconnections may be too low to efficiently advance AI deployment. The Commission should consider raising the threshold to facilities above 100 MW.² If the threshold is too low,

² See, e.g., Initial Comments of the Louisiana and Mississippi Public Service Commissions on ANOPR (Nov. 21, 2025), at pp 8 (proposing that “[t]he threshold should be far higher and set to only address larger loads that are shown to be affected by interconnection delays.”); Comments of Eversource Energy on ANOPR (Nov. 21, 2025), at pp 6 (recommending that “the Commission should consider a voltage or MW threshold (likely much greater than 20 MW) such that transmission providers could still require the largest loads to participate in a federal interconnection process.”); Comments of American Transmission Company LLC on ANOPR (Nov. 21, 2025), at pp 8 (stating that “a 20 MW threshold would be too low and too vague for implementing load interconnection procedures”); Initial Comments of Consumers Energy Company on ANOPR (Nov. 21, 2025), at pp 8-9 (stating that the “proposed 25 MW threshold is

the queue will be bogged down with relatively small interconnections that would better be served through existing load interconnection processes. To ensure that regulatory reforms adequately support efforts to advance AI deployment, streamlined processes should prioritize large facilities above 100 MW. For example, Crusoe has under development facilities in the 1,200 MW range with both grid interconnection and onsite natural gas production. Facilities of this scale will accelerate AI deployment in the United States and should be prioritized through interconnection reforms.

Principle 3: Load and hybrid facilities should be studied together with generating facilities.

Crusoe has revolutionized co-located generation and scaled the technology to enable colocation to high levels currently deployed today. Consistent with this third principle, hybrid facilities combining interconnection requests with onsite generation should be prioritized as they can reduce load requirements while potentially supporting the grid during peak demand.³ Joint interconnection studies for hybrid facilities that bring new generation with load will expedite the interconnection process and minimize the need for costly and time-consuming construction of network upgrades because the impact of the new load can be offset by the addition of new generation.

too low” and a “100 MW nationwide threshold would be more appropriate than the ANOPR’s 20 MW proposal.”).

³ See, e.g., Comments of Eolian L.P. on ANOPR (Nov. 21, 2025), at pp 25 (proposing a “streamlined, single study, serial interconnection process...for hybrid facilities”); Comments of Meta Platforms, Inc. on ANOPR (Nov. 21, 2025), at pp 4-5 (strongly supporting “expedited review of facilities that co-locate large loads and generation”); Comment of the Information Technology Industry Council on ANOPR (Nov. 21, 2025), at pp 1-2 (stating that the “bundling of new generation and load in interconnection studies would streamline efficiencies and necessitate fewer system upgrades.”).

Principle 4: Load and hybrid facility interconnection requests should be subject to heightened standards to prevent speculative interconnection requests from clogging the queue.

Speculative interconnection requests are hindering AI deployment across the country. Speculative requests will continue to delay interconnection for large scale projects and lead to uncertainty in real load demand. In addition to causing delays in interconnecting non-speculative load, speculative interconnection requests are problematic in a number of other areas such as ensuring resource adequacy, transmission planning, and reliability.⁴ Crusoe supports the comments of Google⁵ and OpenAI⁶ asking the Commission to impose heightened standards for interconnection requests, including significant increase in deposit fees and site control requirements to deter speculative requests. Heightened interconnection standards will also provide transmission planners with more accurate information to expedite interconnection studies and get a clearer picture of a new large load's impact on the transmission system.

Interconnection queue management and technical requirements must balance speed with legitimacy. Crusoe supports the ANOPR's goal of significantly expediting interconnection service for large loads. To do this, the Commission should seriously consider whether managing an interconnection queue on a first-come, first-served serial (or cluster) basis will

⁴ For example, the load forecast in ERCOT is difficult to determine with accuracy because of the potential existence of speculative load interconnection requests. Currently, ERCOT has a peak load of approximately 86 GW. Utilities in ERCOT have forecasted the peak to be as high as 218 GW by 2031, but ERCOT has adjusted the load forecast to 148 GW by 2031 to remove speculative load interconnection requests.

<https://interchange.puc.texas.gov/search/documents/?controlNumber=55999&itemNumber=121>; <https://www.ercot.com/calendar/04072025-Board-of-Directors-Meeting>. Sharp differences in load forecasts can be minimized by preventing speculative load interconnection requests. The number of GW currently in the ERCOT large load interconnection queue has quadrupled since last year. <https://www.latitudemedia.com/news/ercots-large-load-queue-has-nearly-quadrupled-in-a-single-year/>.

⁵ Comments of Google LLC on ANOPR (Nov. 21, 2025), at pp 12-13.

⁶ Initial Comments of OpenAI Inc. on ANOPR (Nov. 21, 2025), at pp 3.

achieve the goal of interconnecting large loads in an efficient and expedited manner. Serial queues with low thresholds will not be workable. The Commission should instead establish clear, objective criteria that require interconnection customers to demonstrate both technical capability and financial commitment beyond mere qualitative assurances of curtailment flexibility. Without such safeguards, speculative applicants could secure advantageous queue positions without demonstrating actual technical ability to meet the enhanced performance requirements, thereby creating barriers to entry for legitimate developers and introducing significant uncertainty into project valuation and transferability.

This dynamic would have particularly adverse consequences for the financial and investment markets where the ability to conduct meaningful due diligence on the enforceability and legitimacy of interconnection rights is essential to facilitating efficient capital deployment in the data center and energy infrastructure sectors. Crusoe therefore urges the Commission to develop robust qualification standards that balance the need for expedited processing with the imperative of maintaining queue integrity and market confidence. Such qualification standards can include site control requirements,⁷ minimum queue fees of \$100,000 or greater, and tendering full security for transmission upgrades.

Principle 5: Hybrid facilities should be studied based on the amount of injection or withdrawal rights.

New large load and hybrid facility interconnection processes should study the specific load withdrawal and generation injection rights requested by large load and hybrid facility developers. Hybrid facility structures cause significantly less impact on the transmission grid than large loads that do not bring generation. Thus, where hybrid facilities are proposed, the

⁷ Options to purchase should be acceptable to demonstrate site control to enable developers to ensure power supplies are available before committing to purchases or leases.

load and generation should be studied together to get an accurate picture of the load's impact on the grid.⁸ The Commission should recognize that a one-size-fits-all study process that does not distinguish between hybrid facilities and large loads will discourage innovative solutions to expediting AI computing technologies.

The Commission should follow ERCOT's lead in exploring the feasibility of an expedited study process for controllable load and generation to interconnect and economically dispatch in the wholesale market and to participate as voluntary load shed resources. Hybrid facilities should be viewed as flexible resources that can provide system-wide reliability benefits while having a minimal impact on the transmission system.

Finally, Crusoe supports SPP's currently proposed High Impact Large Load (HILL) and High Impact Large Load Generation Assessment (HILLGA) model because it is proactive and supported by a broad group of industry stakeholders. SPP's model recognizes the need to preserve grid-connected generation for resource adequacy and reliability concerns.

Principle 6: Hybrid interconnections should be required to install protection facilities to prevent unauthorized injections or withdrawals.

The Commission should require hybrid interconnections to install the system protection facilities necessary to prevent unauthorized injections or withdrawals that exceed the respective

⁸ See, e.g., Motion to Intervene and Comments of DTE Electric Company on ANOPR (Nov. 21, 2025), at pp 11 (stating that “[b]y studying large loads and their supporting generation together, transmission providers can better manage the challenges posed by rapid load growth, avoid unnecessary regulatory complexity, and deliver faster, more predictable outcomes for all stakeholders.”); Meta Platforms, Inc., Comments at pp 5 (agreeing that large load should “be studied together with generating facilities”); Comments of the Pennsylvania Office of Consumer Advocate and Delaware Division of the Public Advocate on ANOPR (Nov. 21, 2025), at pp 20 (agreeing “that hybrid facilities should be studied based on the amount of injection and/or withdrawal rights requested, so long as the interconnection study process is comprehensive, technically correct, and aligns with agreed upon operational expectations that will be executed.”).

rights available to the interconnection customer.⁹ Minimum telemetry, protection standards and operating schemes should be established and strictly enforced.

Principle 7: Interconnection studies should be expedited for large loads that are curtailable and co-located generators that are dispatchable.

Crusoe supports flexible interconnection rules that will streamline large load interconnections for prioritized national projects. Specifically, Crusoe supports expedited interconnection study procedures to prioritize proven flexible load technologies. As a leader in flexible computing, Crusoe supports expedited processes as discussed by Emerald AI,¹⁰ Google,¹¹ OpenAI,¹² and others.¹³ However, it is critical that companies must demonstrate actual technical ability to curtail loads and advance flexible computing to be eligible for expedited interconnection. If there are no stringent requirements for flexible load, the Commission could inadvertently create inefficiencies if unproven technologies are prioritized

⁹ See, e.g., Comments of the Solar Energy Industries Association on ANOPR (Nov. 21, 2025), at pp 15 (generally supporting the recommendation that “the large load and co-located facilities install the system protection facilities necessary to prevent unauthorized injections or withdrawals”); Comments of Critical Loop on ANOPR (Nov. 14, 2025), at pp 5 (recommending “[m]inimum controls and telemetry for LLFC participants”); Comments of Transmission Access Policy Study Group on ANOPR (Nov. 21, 2025), at pp 16 (stating “it is essential that they be required to have system protection facilities/mechanisms/schemes that are adequate to enforce these limits”).

¹⁰ Comments of Emerald AI on ANOPR (Nov. 21, 2025), at pp 1 (“We are particularly supportive of Principle 7 and urge the Commission to create a ‘Flexible Load Fast Track’ (FLFT)”).

¹¹ Google LLC, Comments at pp 11 (“The Commission Should Consider Expedited Interconnection Treatment of Voluntarily Flexible Loads”).

¹² OpenAI Inc., Initial Comments at pp 2 (“Consistent with Principle 7 of the DOE ANOPR, large loads that agree to be curtailable should be expedited.”).

¹³ See Comments of Digital Power Network on ANOPR (Nov. 20, 2025), at pp 4 (stating that “[t]his reform is essential” and “[t]here also needs to be an even higher priority for company submissions that have proven their ability to curtail/dispatched at their facilities vs. new entries.”); Comments of ENGIE North America, Inc. on ANOPR (Nov. 24, 2025), at pp 9 (supporting “the concept of expedited interconnection for these types of facilities where load brings its own generation and/or storage.”); Comments of International Energy Credit Association on ANOPR (Nov. 21, 2025), at pp 11-12 (supporting the use of ‘Controllable Dispatch’ and other technological means of managing outages or derates of either a Data Center or its associated generation facility.”).

and expedited. By offering expedited interconnection schedules for demonstrated flexible load, the Commission will encourage innovative solutions to ensure flexible load can perform as expected. For example, any AI computing facility can become a fully flexible load by switching to their own 100% back up power source, including a natural gas plant or by pairing wind or solar with batteries.

Once in the study process, subsequent rule or process changes should not apply to pending interconnection requests retroactively. Crusoe is concerned that projects can get caught in a never-ending spiral of rule changes that will cause considerable delays even for projects that are nearly finished with the interconnection process.¹⁴ Crusoe has been an industry leader in developing new technology, such as voltage ride-through (VRT) capability solutions, but such innovations have caused additional unforeseen delays in receiving approvals to interconnect. New rules when applied retroactively will only cause delays and clog the interconnection queues with the inevitable re-studies. Any new rules developed in this NOPR process or thereafter should apply to new projects and should not move the goal posts for projects already being processed under an earlier set of rules.

Principles 8, 11 and 12: Load and hybrid facilities should be responsible for transmission and ancillary services costs that are fair and reasonable.

Large loads and hybrid facilities should pay their fair share of current and future grid costs based on their use of the grid. For load, that means the power that is withdrawn from the grid, not what the load consumes in total.

¹⁴ For example, Senate Bill 6 requires ERCOT to conduct additional studies for new large loads to co-locate with existing generation resources, a requirement that is now delaying projects that were already past the study process. *See* S.B. 6, 89th Leg. (Tex. 2025). In addition, ERCOT has identified voltage ride through (VRT) capability difficulties, yet there are no standards that exist for measuring load VRT capability. This is resulting in uncertainty and delay in the interconnection process.

Transmission and ancillary services charges should be assigned to hybrid facilities based on the specific grid services that the load and generation uses. Under ERCOT's generation netting rules, for example, generation and load behind a shared interconnection are netted together such that in each market settlement interval (fifteen minutes) if more power is flowing out (net generation on the grid) then it is settled as generation and no grid charges apply. If more power is flowing in during the interval (net load), then it is settled as load with applicable grid charges. Large loads and hybrid facilities should thus pay their fair share of current and future grid costs based on their use of the grid.

Principle 9: Load and hybrid facility interconnection customers should have the option to build interconnection facilities and network upgrades.

Load and hybrid facility interconnection customers should have the same option to build interconnection facilities and network upgrades as currently provided to generator interconnection customers.¹⁵ In particular, co-located generation in a hybrid facility interconnection structure is in a good position to expedite the construction of eligible upgrades.

Principle 10: An existing generating facility that seeks to enter a partial suspension to serve a new load at the same location must go through a system support resource (SSR) or reliability must run type study.

The Commission should first define what an "existing generator" is for purposes of the necessary withdrawal studies. For example, an existing generator could be defined as any

¹⁵ See Comments of Advanced Energy United on ANOPR (Nov. 21, 2025), at pp 25 (supporting "this as a critical policy to support competition and give load customers opportunities to speed and lower the cost of building network upgrades"); Comments of Arevia Power on ANOPR (Nov. 21, 2025), at pp 2 (stating that it "would be open and supportive to this priority."); Comments of Constellation Energy Generation, LLC on ANOPR (Nov. 21, 2025), at pp 26-27 (agreeing "that the option to build should be provided to large load customers subject to host utility standards as it is currently provided to generator interconnection customers."); Motion to Intervene and Comments of Tri State Generation and Transmission Association, Inc. on ANOPR (Nov. 21, 2025), at pp 8 (supporting "the interconnection customer's right to build, as long as the same rules apply to the transmission owner and the interconnection customer.").

generator that was operational as of January 1, 2026. That operational date should remain the same and not become a moving target. Any generation installed after that date would not be considered an existing generator that is subject to withdrawal studies.

Any proposal to withdraw an existing generator from market availability should be subject to a transmission security study and a resource adequacy study, with a defined timeline for each. If operational problems are shown in the studies, RTOs should be able to impose conditions on the withdrawing generator, such as giving the RTO the right to call back the power to the grid when needed.

Principle 13: There must be a plan to implement large load and hybrid interconnection reforms.

Given the current data center demand, Crusoe supports the DOE Secretary's request for final Commission action by April 30, 2026. As noted above, any new rules developed should not apply to interconnection requests retroactively. Any new rules developed in this rulemaking process or thereafter should apply to new projects and should not move the goal posts for projects already being processed under an earlier set of rules.

B. Other Needed Reforms to Support Rapid Deployment of AI Infrastructure.

1. Coordinated federal action is needed to improve transmission construction and equipment procurement timelines.

Crusoe commends the proposed ANOPR's policy improvements to shorten study timelines for large load and hybrid facility interconnections. That is one of the major sources of delay in the deployment of AI computing infrastructure. Two other sources of delay are (1) transmission inadequacy and construction timelines and (2) the long lead times for transmission and generation equipment. While these sources of delay are not specifically addressed in the ANOPR, it is imperative that federal policies step in to address these delays on a

comprehensive basis. For example, to address transmission construction delays and lack of interregional planning, the Commission should be more aggressive in deploying its backstop siting authority where needed.

In addition, supply chain constraints for gas and electrical long lead equipment represent a separate challenge that will require complementary policy solutions. Even with expedited interconnection studies, the time required to manufacture, deliver, and energize network upgrades and new transmission facilities will continue to create long lead times before large loads can be energized. Currently, developers of transmission, generation and load interconnections are all competing for the same equipment. Interconnections can require new, custom-built transformers and voltage step-down equipment, which, in some cases, require up to 48 months to procure. This procurement timeline effectively becomes the critical path for project development, regardless of how fast the interconnection study process is conducted.

Thus, to fully realize the benefits of expedited large load interconnection procedures, the Commission and other federal agencies should work together to develop complementary federal policy initiatives focused on domestic manufacturing capacity and production timelines for electrical long lead equipment that could significantly enhance the rapid deployment of data centers and other large loads. Strengthening the domestic supply chain for transformers, switchgear, and other critical electrical equipment would not only accelerate large load interconnection timelines but also enhance energy security and grid resilience more broadly.

2. The Commission should adopt PUHCA reforms to remove federal impediments to power supply arrangements by eliminating “sleeve” PPA requirements where retail sales are allowed by states.

Crusoe supports EDF power solution, Inc.'s recommendations regarding reforms to the Commission's Public Utility Holding Company Act (PUHCA) regulations that effectively

require independent power producers to enter into sleeve PPAs to maintain PUHCA exemptions.¹⁶ By requiring sleeve PPAs in states that would otherwise allow retail sales to large loads, the PUHCA regulations create significant impediments to hybrid facility development and increase costs. Sleeve requirements impose unnecessary transaction costs and delays by requiring a regulatory "middleman" whose involvement adds no operational or commercial value but is necessary solely to maintain PUHCA exemptions for upstream owners and investors. Eliminating the sleeve requirement to maintain PUHCA exemptions for upstream owners would reduce costs and accelerate project development.

III. CONCLUSION

For the foregoing reasons, Crusoe requests that the Commission proceed expeditiously to develop interconnection rules that ensure the reliable and speedy interconnection of new large loads and hybrid facilities of national importance.

Respectfully submitted,

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¹⁶ Comments of EDF Power Solutions, Inc. on ANOPR (Nov. 21, 2025), at pp 5 (recommending PUHCA reform).